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Extended incremental nonlinear dynamic inversion for optical flow control of micro air vehicles

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ABSTRACT

Optical flow-based control strategies have always inspired robotic scientists, especially those in the field of Micro Air Vehicles (MAVs), thanks to their computational efficiency and relative simplicity. A major problem is that the success of optical flow control is governed by the availability of distance estimates, while optical flow provides only the ratio of velocity to distance. Therefore, with only monocular visual information, the inherent nonlinearity of optical flow observables has imposed several challenges in the controller design. In this paper, we propose a newly formulated controller, Extended Incremental Nonlinear Dynamic Inversion (EINDI), to deal with nonlinearities in the system output, such as optical flow control problems. The proposed method unlocks the potential of its predecessor (INDI) in output feedback control by removing the common assumption of time-scale separation, allowing internal dynamics to exist, and requiring only the input and output measurements. Furthermore, the EINDI method has been implemented on an MAV and tested successfully for optical flow landing in a simulation and a real-world outdoor environment. Both simulation and flight test results show 1) good tracking performance of the EINDI control compared to the conventional feedback control, 2) smooth landing trajectories without any oscillation, and 3) fast adaptation of the EINDI control even for landings at different heights and desired setpoints.

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1. Introduction

In recent years, small flying robots or Micro Air Vehicles (MAVs) have gained a huge interest in robotics research and applications due to their capability of flying in narrow environments and close to humans [1–3]. Yet, these vehicles are facing enormous challenges in flying autonomously with severely limited computational power and payload capability. On the other hand, tiny flying insects perform complex tasks in unfamiliar environments very well, despite very little neural and sensory resources [4,5]. These insects primarily rely on their eyes and have incredibly efficient and robust solutions to undertake these perception and control problems [6]. These solutions can serve as the bio-inspired design principles for MAV autonomous control strategies.

The main visual cue used by flying insects to know their movement and to sense objects in the surroundings while flying is optical flow. Optical flow refers to the apparent motion objects in a scene perceived by an observer during their movement through the world [7]. Flying insects heavily rely on optical flow to directly control their maneuvers. For instance, honeybees use optical flow to keep a safe clearance from the surrounding objects and control their speeds when they are moving through narrow spaces [8]. They also carry out landings by keeping optical flow constant, which exponentially decreases both their velocity and height to almost zero at the same time [6,9]. One of the optical flow cues which has first been found to have a significant contribution to the insects' landing control is ventral flow (i.e., the ratio of the lateral velocity and height). This strategy has been studied and implemented on MAVs and space vehicles for autonomous landings [4,10–12]. In fact, the success of these landings is governed by a pitching law profile, which requires the estimates of the height and vertical velocity of these vehicles [11,13]. Hence, another approach using constant flow divergence (i.e., the ratio of the vertical velocity and height) or decreasing time-to-contact (i.e., the reciprocal of the flow divergence) has recently been studied [14–17]. A series of experiments show evidence that honeybees also land with constant flow divergence [16].

Optical flow strategies followed by flying insects are remarkable, as they seem to guarantee a smooth landing by only using

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optical flow. The key to this success lies in its scaleless property, which provides the ratio of velocity to distance but not velocity and distance separately [18]. In other words, these insects land without any knowledge of velocity and distance but the visual cues representing their ratio. Using these visual cues would be contrary to typical engineering solutions that opt to separate these physical quantities [19,20]. Thus, it imposes several challenges on designing the controller that uses optical flow for MAV landings. First, recent studies showed that the conventional linear controllers which guarantee a desired optical flow landing trajectory are height-dependent [21,22]. For instance, the stable control gains in proportional (P) [21] and proportional-integral (PI) [22] controllers are a function of heights. A similar observation was found in decreasing time-to-contact strategy, where the control gains are scheduled based on the various formulation of the initial time-to-contact [23]. Second, without using additional sensory resources, some information, which animals or humans use to predict their motion based on the actions, i.e., efference copies, are utilized in MAVs as the control signal to estimate heights [24] or to design adaptive controllers [21,22]. Third, the optical flow, perceived as the output of the MAV system, is a highly nonlinear function of the system states [25]. Most approaches use a simple Proportional-Integral-Derivative (PID) controller to regulate the optical flow to a desired constant. However, the stability and performance of using such linear controllers on a nonlinear system can be questionable as they may only work at specific operating points, such as within a certain range of heights.

Alternatively, nonlinear and adaptive control methods could be used to deal with the system nonlinearities without gain scheduling [26–29]. Among them, Nonlinear Dynamic Inversion (NDI) is a model-based controller that establishes the direct relationship between the output and input, designs a virtual input with linear methods, and solves for the physical input by inversion. This method has been applied to a wide range of flight models [30–36]. However, the NDI method assumes that the model is completely accurate, which is highly unlikely to happen for most systems in real life due to model simplification and external disturbances.

To this end, an incremental form of NDI, or so-called INDI, is introduced to desensitize the model inaccuracy with sensor measurements in flight control [37,38]. The closed-loop control of feeding back accurate angular accelerations eliminated all non-input dependent uncertainties and showed excellent performance in the presence of an aerodynamic model mismatch. However, in practice, angular accelerations are not readily available in most standard systems. Several attempts have been made to estimate angular accelerations from angular rate measurements, which are available in an Inertial Measurement Unit (IMU), and to solve the undesired time-delay consequences [39–41]. These INDI approaches are beneficial, especially for the inner-loop or attitude control which is highly influenced by the complex aerodynamic model. Besides, some unconventional types of MAVs which possess different flight modes, i.e., hover, transition, and forward flight, used INDI controllers for attitude (inner-loop) control [42,43] and position (outer-loop) control [44] to deal with coupling effects in dynamic models. The INDI control for nonlinear systems is typically praised, as it seems to enable model reduction and nonlinear control practicality. Recently, a detailed stability analysis of a closed-loop system with INDI control was made using Lyapunov methods and nonlinear system perturbation theory [45]. It was shown that INDI itself is more robust than NDI in terms of disturbance rejection and model uncertainties.

A major problem is that existing INDI control laws assume that 1) the validity of the time-scale separation principle in both control law design and stability analysis, and 2) the full state measurements are available. Both assumptions restrict the applications of INDI controllers for output feedback control. Firstly, time-scale separation is not always applicable, especially for measurements with unbounded timescales. For example, in the case of flow divergence landings, when the distance is very small, the change of flow divergence will be very fast and theoretically unbounded. Therefore, the flow divergence landing process does not accept the assumption of time-scale separation. Secondly, obtaining full states in MAV systems may need additional sensors or heavy computations. This is unfavorable in real practices, especially for MAVs, in which both weight and computation are severely constrained.

The **main contribution** of this article is that we completely exclude the time-scale separation assumption for the first time and only require output measurements in our newly formulated EINDI control law to generalize it to most cases even with internal dynamics and output nonlinearities. Thus, it can be used in optical flow control to deal with the nonlinearity or inherent scaleless property of optical flow, without the need of estimating or measuring distances through additional sensors.

The remainder of this paper is structured as follows. Section 2 introduces the strategy of the smooth landing with constant flow divergence. Section 3 presents the concept of NDI and INDI methods and the formulation of the EINDI method for flow divergence landings with simulation results. Then, section 4 validates the EINDI method with only output measurements on real-world flight tests and compares the results with that of conventional feedback control. Lastly, section 5 shows the advantages of the proposed EINDI and addresses the challenges and possibilities for future research.

2. Constant flow divergence landing

First consider Fig. 1, which defines the body reference frame of an MAV ($O^bX^bY^bZ^b$) and world reference frame ($O^wX^wY^wZ^w$). A camera is attached to the body of the MAV and looking downward. Thus, the camera reference frame ($O^cX^cY^cZ^c$) is assumed to align with the body reference frame. During landing, the optical flow captured from the camera show a divergent pattern. We define this observation as flow divergence. The flow divergence ϑ can be formulated as the ratio of the vertical velocity v to the height above ground d:

$$\vartheta = \frac{v}{d}. \tag{1}$$

When the MAV is descending, v is negative while d is always positive, and thus, ϑ will be negative.

The constant flow divergence approach is used to control the vertical dynamics of MAVs by keeping the flow divergence constant, e.g., $\vartheta = \vartheta^*$ for landing, where ϑ^* is a negative constant. As a result, the time response of the height d, vertical velocity v, and vertical acceleration a can be estimated as follows:

$$d(t) = d_0 e^{\vartheta^* t}, \tag{2}$$

$$v(t) = \vartheta^* d_0 e^{\vartheta^* t}, \tag{3}$$

$$a(t) = \vartheta^{*2} d_0 e^{\vartheta^* t}, \tag{4}$$

where d_0 is the initial height above the landing surface. As $t \to \infty$, all these states decrease exponentially to zero.

3. Extended incremental nonlinear dynamic inversion control

Nonlinear Dynamic Inversion (NDI) is a control method for nonlinear systems based on their models. The idea is to 1) find the direct relation between the output and the input by differentiation, 2) design a virtual control input using linear control methods, and then 3) solve for the physical control input by inversion. The INDI method and the proposed EINDI method are developed

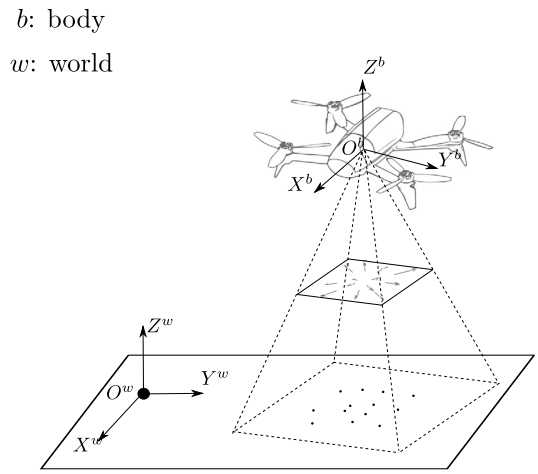

Fig. 1. MAV body ($O^bX^bY^bZ^b$) and world ($O^wX^wY^wZ^w$) reference frames. O^b is fixed at the center of gravity of the MAV, X^b points forward, Y^b is starboard, and Z^b points upward. The world reference frame is located on the ground and follows the North-East-Up ($X^w - Y^w - Z^w$) system.

based on the concept of NDI, but depend less on system models. Instead, they identify an (extended) incremental model from the sensor measurements and calculate the input increment by inversion. Therefore, this section starts with the NDI method and its control law formulation to the flow divergence landing with full state measurements in subsection 3.1. Then, subsection 3.2 describes the incremental models and the INDI method with full state measurements. Subsection 3.3 proposes the EINDI method to deal with nonlinear control problems with internal dynamics and only output measurements. Subsection 3.4 applies the EINDI method to the flow divergence landing problem and generates the control law. And lastly, subsection 3.5 validated the control method with simulated landing experiments.

3.1. Nonlinear dynamic inversion

Nonlinear input-affine systems can be generally written as:

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u}, \tag{5}$$

$$\boldsymbol{y} = \boldsymbol{h}(\boldsymbol{x}), \tag{6}$$

where $\boldsymbol{x} \in \mathcal{R}^n$ is the system state, $\boldsymbol{u} \in \mathcal{R}^m$ is the system input, $\boldsymbol{y} \in \mathcal{R}^p$ is the system output, $\boldsymbol{f} : \mathcal{R}^n \to \mathcal{R}^n$ and $\boldsymbol{h} : \mathcal{R}^n \to \mathcal{R}^p$ are smooth vector fields, and $\boldsymbol{g} : \mathcal{R}^n \to \mathcal{R}^{n\times m}$, whose columns are smooth vector fields. If $p < m$ or $p > m$, this problem becomes an over-determined or under-determined control problem, and control allocation techniques or weighted least square methods can be used to solve those problems, respectively. The rest paper assumes that the numbers of inputs and outputs are the same, $p = m$, for both derivations and applications.

The first step of NDI is input-output linearization by differentiating the output until the input-affine form appears [46,39,45,41]. The first-order derivative of the lth output is

$$\begin{aligned}\frac{dh_l(\boldsymbol{x})}{dt} &= \frac{\partial h_l(\boldsymbol{x})}{\partial \boldsymbol{x}}\frac{d\boldsymbol{x}}{dt} = \nabla h_l(\boldsymbol{x})[\boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u}] \\ &= L_f h_l(\boldsymbol{x}) + L_g h_l(\boldsymbol{x})\boldsymbol{u},\end{aligned} \tag{7}$$

where ∇ is Jacobian operator, and $L_f h_l(\boldsymbol{x}) = \nabla h_l(\boldsymbol{x})\boldsymbol{f}(\boldsymbol{x})$ is first-order Lie derivative along the vector field $\boldsymbol{f}(\boldsymbol{x})$ [47]. If the term $L_g h_l(\boldsymbol{x})$ is non-zero, the first-order differentiation of the output $\dot{\boldsymbol{y}}$ is input-affine, as follows:

$$\dot{y}_l = L_f h_l(\boldsymbol{x}) + L_g h_l(\boldsymbol{x})\boldsymbol{u}. \tag{8}$$

Otherwise, the first-order differentiation of the lth output $\dot{y}_l$ is non-affine in input and will be differentiated until the Lie derivative w.r.t. $\boldsymbol{g}(\boldsymbol{x})$ is nonzero:

$$\begin{aligned}y_l^{(r_l)} &= L_f^{r_l} h_l(\boldsymbol{x}) + L_g L_f^{r_l-1} h_l(\boldsymbol{x})\boldsymbol{u} \\ &= L_f^{r_l} h_l(\boldsymbol{x}) + \left[L_{g_1} L_f^{r_l-1} h_l(\boldsymbol{x}) \cdots L_{g_m} L_f^{r_l-1} h_l(\boldsymbol{x})\right]\boldsymbol{u},\end{aligned} \tag{9}$$

where $\boldsymbol{g}_1, \cdots, \boldsymbol{g}_m$ are the smooth column vector fields of $\boldsymbol{g}$.

For each output, $r_l \geq 1$, and the total relative degree $r = r_1 + r_2 + \ldots + r_p$ is no more than the number of the state n. Therefore, the system has the total relative degree $r \in [p, n]$. If $r = n$, the first $r_l - 1$ derivatives for each output will transform the original x coordinate system into the new ξ coordinate system with the normal form, as follows:

$$\boldsymbol{\xi} = \begin{bmatrix} \xi_1 \\ \vdots \\ \xi_{r_1} \\ \xi_{r_1+1} \\ \vdots \\ \xi_r \end{bmatrix} = \boldsymbol{T}_1(\boldsymbol{x}) = \begin{bmatrix} L_f^0 h_1(\boldsymbol{x}) \\ \vdots \\ L_f^{r_1-1} h_1(\boldsymbol{x}) \\ L_f^0 h_2(\boldsymbol{x}) \\ \vdots \\ L_f^{r_p-1} h_p(\boldsymbol{x}) \end{bmatrix}. \tag{10}$$

As the new system states $\boldsymbol{\xi}$ are smooth functions of $\boldsymbol{x}$ and invertible, this transformation is a diffeomorphism in the defined domain. And the new system of $\boldsymbol{\xi}$ can be written in the controllable canonical form as below:

$$\begin{bmatrix} \dot{\xi}_1 \\ \dot{\xi}_2 \\ \vdots \\ \dot{\xi}_{r_1} \\ \dot{\xi}_{r_1+1} \\ \vdots \\ \dot{\xi}_r \end{bmatrix} = \begin{bmatrix} \xi_2 \\ \xi_3 \\ \vdots \\ L_f^{r_1} h_1(\boldsymbol{x}) \\ \xi_{r_1+2} \\ \vdots \\ L_f^{r_p} h_p(\boldsymbol{x}) \end{bmatrix} + \begin{bmatrix} 0 & & \\ 0 & & \\ \vdots & & \\ L_{g_1} L_f^{r_1-1} h_1(\boldsymbol{x}) & \cdots & L_{g_m} L_f^{r_1-1} h_1(\boldsymbol{x}) \\ 0 & & \\ \vdots & & \\ L_{g_1} L_f^{r_p-1} h_p(\boldsymbol{x}) & \cdots & L_{g_m} L_f^{r_p-1} h_p(\boldsymbol{x}) \end{bmatrix}\boldsymbol{u}. \tag{11}$$

On the other hand, $r < n$ implies that there are $n - r$ internal dynamics, which are not directly observable from the input-output linearization. And it is always possible to find $n - r$ more functions

$$\boldsymbol{T}_2(\boldsymbol{x}) = \begin{bmatrix} T_{r+1}(\boldsymbol{x}) \\ \vdots \\ T_n(\boldsymbol{x}) \end{bmatrix}, \tag{12}$$

so that the transformation $\boldsymbol{T} = [\boldsymbol{T}_1; \boldsymbol{T}_2]$ in the defined domain is a diffeomorphism [46]. The additional functions can be chosen in the defined domain in such a way that

$$L_{g_j} T_i(\boldsymbol{x}) = 0 \quad for\ all \quad i \in [r+1, n],\ j \in [1, m]. \tag{13}$$

And the transformation to the new z coordinate system can be defined as

$$\boldsymbol{z} = \begin{bmatrix} \boldsymbol{\xi} \\ \boldsymbol{\eta} \end{bmatrix} = \boldsymbol{T}(\boldsymbol{x}) = \begin{bmatrix} \boldsymbol{T}_1(\boldsymbol{x}) \\ \boldsymbol{T}_2(\boldsymbol{x}) \end{bmatrix}. \tag{14}$$

The output $\boldsymbol{y}$ is linear to the new state:

$$\boldsymbol{y} = \begin{bmatrix} y_1 \; y_2 \; y_3 \; \cdots \; y_p \end{bmatrix}^T = \begin{bmatrix} \xi_1 \; \xi_{r_1+1} \; \xi_{r_1+r_2+1} \; \cdots \; \xi_{r-r_p+1} \end{bmatrix}^T. \tag{15}$$

And the linearized input-output relation can be simply written as

$$\boldsymbol{y}^{(\boldsymbol{r})} = L_f^{\boldsymbol{r}} \boldsymbol{h}(\boldsymbol{x}) + L_g L_f^{\boldsymbol{r}-1} \boldsymbol{h}(\boldsymbol{x}) \boldsymbol{u}. \tag{16}$$

To solve the above nonlinear control problem, the virtual control input is introduced to separate the system into a linear outer loop

$$\boldsymbol{y}^{(\boldsymbol{r})} = \boldsymbol{v}, \tag{17}$$

and a nonlinear inner loop

$$\boldsymbol{v} = L_f^{\boldsymbol{r}} \boldsymbol{h}(\boldsymbol{x}) + L_g L_f^{\boldsymbol{r}-1} \boldsymbol{h}(\boldsymbol{x}) \boldsymbol{u}. \tag{18}$$

The virtual input $\boldsymbol{v}$ can be solved using linear feedback control techniques. And the physical control input $\boldsymbol{u}$ can be solved by inverting Eq. (18) as follows:

$$\boldsymbol{u} = L_g L_f^{\boldsymbol{r}-1} \boldsymbol{h}(\boldsymbol{x})^{-1} [\boldsymbol{v} - L_f^{\boldsymbol{r}} \boldsymbol{h}(\boldsymbol{x})]. \tag{19}$$

Considering the vertical landing of an MAV equipped with a single camera looking downwards, the system state consists of the height d and the vertical velocity v, $\boldsymbol{x} = [d, v]^T$, and the input is assumed to be the vertical acceleration, $u = a$. Therefore, the equations of motion can be written in the notation of a state-space model as Eq. (5):

$$\begin{bmatrix} \dot{d} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} v \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} a, \tag{20}$$

where $\boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} v \\ 0 \end{bmatrix}$ and $\boldsymbol{g}(\boldsymbol{x}) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.
The system output is the observed flow divergence ϑ, $y = \vartheta$:

$$y = h(\boldsymbol{x}) = \frac{v}{d}, \tag{21}$$

which is a nonlinear function of the states.

The Lie derivative of $h(\boldsymbol{x})$ along the vector field $\boldsymbol{f}(\boldsymbol{x})$ and $\boldsymbol{g}(\boldsymbol{x})$ are $L_f h(\boldsymbol{x}) = -v^2 d^{-2}$ and $L_g h(\boldsymbol{x}) = d^{-1}$, which is non-zero. Therefore, the first-order derivative of the output can be used to apply the NDI control:

$$\dot{y} = -\frac{v^2}{d^2} + \frac{1}{d} a = -y^2 + \frac{1}{d} a. \tag{22}$$

The virtual control input $\nu = \dot{y}$ is designed to form a stable outer linear system with a constant desired flow divergence ϑ^*:

$$\nu = \dot{y} = -K_p (y - \vartheta^*), \tag{23}$$

where K_p is the output error feedback gain. The time response of the output is

$$y(t) = (y_0 - \vartheta^*) \cdot e^{-K_p t} + \vartheta^*, \tag{24}$$

where $K_p > 0$ will lead to a stable control of the flow divergence. By inverting Eq. (22), the real control input can be solved:

$$a = d(\nu + y^2). \tag{25}$$

Note that the internal dynamics also have to be stable or bounded for an effective controller. However, it is difficult to evaluate the stability of internal dynamics as it is often nonlinear and coupled with the external dynamics of the system. An alternative is to ensure that all the system states in the original coordinate are stable or bounded. In this vertical landing case, when the MAV lands while keeping a negative flow divergence constant at ϑ^*, the states will be converging as in Eqs. (2)-(3).

The above NDI control law depends on the accurate model of the flow divergence with the perfect assumption of $u = a$, and the only acceleration is generated by the control input. In addition, the NDI control solution depends on the full state feedback, which is not directly measurable with a single camera.

3.2. Incremental nonlinear dynamic inversion

Although the model of the vertical landing can be simplified into Eqs. (20) and (21), the real-world system could never match these equations perfectly. The input is assumed to be the vertical acceleration as a straightforward representation for the equation of motion, which does not depend on other state variables. However, the real input should be an electrical signal, which will be transferred to the total upward thrust and forms the acceleration. This model depends on the mass of the MAV, gravity, actuator dynamics, and even the battery level. Besides, the drag force, wind, and other disturbances will also enlarge the mismatch between the real system dynamics and the simplified model.

Incremental Nonlinear Dynamic Inversion (INDI) can be used to reduce the sensitivity to model mismatches by using an online identified incremental model. INDI, instead of using the system model to find the control input, calculates the control input increment based on a desired virtual input increment. The INDI method can be derived from the system as in Eq. (5) or in a more general nonlinear system:

$$\dot{\boldsymbol{x}} = \overline{\boldsymbol{f}}(\boldsymbol{x}, \boldsymbol{u}), \tag{26}$$

where $\overline{\boldsymbol{f}} : \mathcal{R}^{n+m} \to \mathcal{R}^n$ is a smooth vector field.

The system dynamics around the condition at time t_0 can be linearized approximately by using the first-order Taylor series expansion:

$$\dot{\boldsymbol{x}}(t) \approx \dot{\boldsymbol{x}}(t_0) + F[\boldsymbol{x}(t_0), \boldsymbol{u}(t_0)] \Delta \boldsymbol{x}(t) + G[\boldsymbol{x}(t_0), \boldsymbol{u}(t_0)] \Delta \boldsymbol{u}(t), \tag{27}$$

where $\Delta \boldsymbol{x}(t) = \boldsymbol{x}(t) - \boldsymbol{x}(t_0)$, $\Delta \boldsymbol{u}(t) = \boldsymbol{u}(t) - \boldsymbol{u}(t_0)$, $F[\boldsymbol{x}(t_0), \boldsymbol{u}(t_0)] = \frac{\partial \overline{\boldsymbol{f}}(\boldsymbol{x}, \boldsymbol{u})}{\partial \boldsymbol{x}} |_{\boldsymbol{x}(t_0), \boldsymbol{u}(t_0)} \in \mathcal{R}^{n \times n}$ is the system matrix of the linearized model, $G[\boldsymbol{x}(t_0), \boldsymbol{u}(t_0)] = \frac{\partial \overline{\boldsymbol{f}}(\boldsymbol{x}, \boldsymbol{u})}{\partial \boldsymbol{u}} |_{\boldsymbol{x}(t_0), \boldsymbol{u}(t_0)} \in \mathcal{R}^{n \times m}$ is the control effectiveness matrix, and the second or higher order terms of the elements in vector $[\Delta \boldsymbol{x}(t)^T \;\; \Delta \boldsymbol{u}(t)^T]^T$ have been neglected in this equation.

This equation can be further simplified with a time-scale separation assumption. This assumption holds when the change in control input is considered significantly faster than the change in the system state with high sampling frequency and instantaneous control effects [48]. The fastness of a dynamical state can be analyzed mathematically using singular perturbations [49]. The time-scale separation assumptions have been applied to various types of guidance and control problems in aerospace systems for dimensional reductions [39], stiffness relief [49], and also incremental model simplifications [39,41,48]. For the commonly defined states and control inputs in aerospace systems with sufficiently fast actuator dynamics, the fastness of a state or input dynamics depends on the system order: the lower-order dynamics correspond to a faster time scale, and the higher-order dynamics correspond to a slower timescale [48]. In the vertical landing example, the control input, which is the vertical acceleration, a, has more effect on the state changes compared to the changes in aerodynamic force and procession moment due to changes in the vertical velocity v or the height d, which can be proved with singularly perturbation theory (see Appendix A).

When the time-scale separation principle is applicable, we will have $F[\boldsymbol{x}(t_0), \boldsymbol{u}(t_0)]\Delta\boldsymbol{x}(t) \ll G[\boldsymbol{x}(t_0), \boldsymbol{u}(t_0)]\Delta\boldsymbol{u}(t)$, and Eq. (27) can be simplified into

$$\dot{\boldsymbol{x}}(t) \approx \dot{\boldsymbol{x}}(t_0) + G[\boldsymbol{x}(t_0), \boldsymbol{u}(t_0)]\Delta\boldsymbol{u}(t). \tag{28}$$

Replacing the differentiated state by the virtual input $\dot{\boldsymbol{x}} = \boldsymbol{v}$, the control increment can be found by inverting Eq. (28):

$$\Delta\boldsymbol{u}(t) = G[\boldsymbol{x}(t_0), \boldsymbol{u}(t_0)]^{-1}[\boldsymbol{v} - \dot{\boldsymbol{x}}(t_0)], \tag{29}$$

where the matrix G can be online identified from the measurement or estimation of the differentiated state $\dot{\boldsymbol{x}}$ using Least Square (LS) techniques. The above equation is the INDI control law in most current applications.

The INDI method in Eq. (29) can be used in applications with virtual input $\dot{\boldsymbol{x}} = \boldsymbol{v}$, which indicates 1) it is a full-state feedback problem $\boldsymbol{y} = \boldsymbol{x}$ and $n = p = m$, and 2) the relative degree for each output equals to 1, $r_l = 1$. The INDI method can also be used in systems where $\boldsymbol{y} \neq \boldsymbol{x}$ when the relative degree of the system $\boldsymbol{r}$ is known. In other words, the model structure of the input-output linearization as in Eq. (16) is known, but the exact mathematical expression of $L_f^{\boldsymbol{r}}\boldsymbol{h}(\boldsymbol{x})$ and $L_g L_f^{\boldsymbol{r}-1}\boldsymbol{h}(\boldsymbol{x})$ is unknown. Therefore, the input-output relation of the model in Eq. (16) can be generally written as

$$\boldsymbol{y}^{(\boldsymbol{r})} = \boldsymbol{\alpha}(\boldsymbol{x}) + \boldsymbol{\beta}(\boldsymbol{x})\boldsymbol{u}. \tag{30}$$

Taking the first-order Taylor series expansion:

$$\boldsymbol{y}^{(\boldsymbol{r})}(t) \approx \boldsymbol{y}^{(\boldsymbol{r})}(t_0) + F[\boldsymbol{x}(t_0), \boldsymbol{u}(t_0)]\Delta\boldsymbol{x}(t) + G[\boldsymbol{x}(t_0)]\Delta\boldsymbol{u}(t), \tag{31}$$

where $F[\boldsymbol{x}(t_0), \boldsymbol{u}(t_0)] = \frac{\partial[\boldsymbol{\alpha}(\boldsymbol{x})+\boldsymbol{\beta}(\boldsymbol{x})\boldsymbol{u}]}{\partial\boldsymbol{x}}|_{\boldsymbol{x}(t_0),\boldsymbol{u}(t_0)} \in \mathcal{R}^{p\times n}$, and $G[\boldsymbol{x}(t_0)] = \boldsymbol{\beta}(\boldsymbol{x})|_{\boldsymbol{x}(t_0)} \in \mathcal{R}^{p\times m}$.

Similarly, this equation can be further simplified when the time-scale separation principle holds:

$$\boldsymbol{y}^{(\boldsymbol{r})}(t) \approx \boldsymbol{y}^{(\boldsymbol{r})}(t_0) + G[\boldsymbol{x}(t_0)]\Delta\boldsymbol{u}(t). \tag{32}$$

By replacing the differentiated output with the virtual input $\boldsymbol{y}^{(\boldsymbol{r})} = \boldsymbol{v}$, the control increment can be found by inverting Eq. (32):

$$\Delta\boldsymbol{u}(t) = G[\boldsymbol{x}(t_0)]^{-1}[\boldsymbol{v} - \boldsymbol{y}^{(\boldsymbol{r})}(t_0)], \tag{33}$$

where the matrix G can be identified online from the measurement or estimation of $\boldsymbol{y}^{(\boldsymbol{r})}$ using LS techniques. If $r < n$, the internal dynamics also have to be stable or bounded for an effective controller design.

Unlike the NDI control, the INDI approaches use online identified incremental models to cope with the model mismatch, system uncertainty, and other disturbances. In other words, INDI methods are less dependent on the assumption of a perfect model. However, the INDI method in Eq. (29) relies on the assumptions that $\boldsymbol{y} = \boldsymbol{x}$ and that the full states are measurable. When the system structure is partially known, the control increment design in Eq. (33) does not rely on the full state measurements, but it assumes that the time-scale separation principle holds and that $\boldsymbol{y}^{(\boldsymbol{r})}$ can be measured or calculated, which will be very noisy and impractical.

3.3. Extended incremental nonlinear dynamic inversion with only output measurements

Although most current INDI applications have applied the time-scale separation principle [39,41,48], this principle is not applicable to many sensor-based applications, including flow divergence landings, which will be discussed later. In addition, the full state of systems, such as MAVs carrying limited payloads, is often not available. In other words, the control system often needs to control a system without enough information to infer its real states [50,51]. For stochastic systems, the partially observable Markov decision process framework can be applied. For deterministic systems, these types of methods are often referred to as output feedback. The systems still need to be observable, which means that the unmeasurable internal full states can be reconstructed with the observations over a long enough time horizon. For linear systems or nonlinear systems around a linearized operating point, the system is observable when the observability matrix has a full column rank.

The Extended Incremental Nonlinear Dynamic Inversion (EINDI) is proposed to use output feedback methods to deal with nonlinear input-affine systems with internal dynamics and only output measurements. A nonlinear input-affine system Eqs. (5)-(6) in original x coordinate system can be transformed into a new z coordinate system with Eqs. (10)-(14), which is a diffeomorphism in the defined domain. As the state $\boldsymbol{x}$ are not measurable, the EINDI method will be derived in the new z coordinate.

Rewrite the new z coordinate system in Eq. (14) as below:

$$\boldsymbol{z} = \begin{bmatrix} \boldsymbol{\xi} \\ \boldsymbol{\eta} \end{bmatrix} = \boldsymbol{T}(\boldsymbol{x}) = \begin{bmatrix} \boldsymbol{T}_1(\boldsymbol{x}) \\ \boldsymbol{T}_2(\boldsymbol{x}) \end{bmatrix}, \tag{34}$$

where the new state $\boldsymbol{z}$ are smooth functions of $\boldsymbol{x}$ and invertible in the defined domain. The first r states in the new system, $\boldsymbol{\xi}$, are output derivatives as in Eq. (10), and its dynamics can be written in the controllable canonical form as in Eq. (11), or in a more general form

$$\begin{aligned} \dot{\boldsymbol{\xi}} &= A\boldsymbol{\xi} + B[\boldsymbol{\alpha}(\boldsymbol{x}) + \boldsymbol{\beta}(\boldsymbol{x})\boldsymbol{u}] \\ &= A\boldsymbol{\xi} + B[\boldsymbol{\alpha}[\boldsymbol{T}^{-1}(\boldsymbol{z})] + \boldsymbol{\beta}[\boldsymbol{T}^{-1}(\boldsymbol{z})]\boldsymbol{u}], \end{aligned} \tag{35}$$

where A and B are logical matrices representing the canonical form in Eq. (11). Under the condition in Eq. (13), the dynamics of the other $n-r$ states, $\boldsymbol{\eta}$, can be written as

$$\dot{\boldsymbol{\eta}} = [L_f\boldsymbol{T}_2(\boldsymbol{x}) + L_g\boldsymbol{T}_2(\boldsymbol{x})\boldsymbol{u}]|_{\boldsymbol{x}=\boldsymbol{T}^{-1}(\boldsymbol{z})} = L_f\boldsymbol{T}_2(\boldsymbol{x})|_{\boldsymbol{x}=\boldsymbol{T}^{-1}(\boldsymbol{z})}. \tag{36}$$

If $r = n$ and the time-scale separation principle holds, the new states $\boldsymbol{z} = \boldsymbol{\xi}$ are directly observable from the output, and the control increment can be designed similarly to the INDI method in Eq. (33). If $r < n$, the external and internal dynamics of the system are coupled, and the dynamics of the new system state $\boldsymbol{z}$ in Eq. (11) can be generally written as

$$\dot{\boldsymbol{z}} = \boldsymbol{f}_z(\boldsymbol{z}) + \boldsymbol{g}_z(\boldsymbol{z})\boldsymbol{u}, \tag{37}$$

where $\boldsymbol{f}_z : \mathcal{R}^n \to \mathcal{R}^n$ is a smooth vector field, and $\boldsymbol{g}_z : \mathcal{R}^n \to \mathcal{R}^{n\times m}$, whose columns are smooth vector fields. And the output in terms of the new state can be written as

$$\boldsymbol{y} = C\boldsymbol{z}, \tag{38}$$

where $C \in \mathcal{R}^{p\times n}$ is a logical matrix representing linear relation between output and the new state as in Eq. (15).

Although most physical systems are continuous, their output measurements are often discrete. With a sufficiently high, constant data sampling frequency, the nonlinear system in z coordinate can be written in a discrete form:

$$\boldsymbol{z}_{k+1} = \boldsymbol{f}_z(\boldsymbol{z}_k) + \boldsymbol{g}_z(\boldsymbol{z}_k)\boldsymbol{u}_k. \tag{39}$$

The system around $\boldsymbol{z}_k$ can be linearized by using the first-order Taylor series expansion in the new z coordinate:

$$\boldsymbol{z}_{k+1} \approx \boldsymbol{z}_k + F_{k-1}(\boldsymbol{z}_k - \boldsymbol{z}_{k-1}) + G_{k-1}(\boldsymbol{u}_k - \boldsymbol{u}_{k-1}), \tag{40}$$

where $F_{k-1} = \frac{\partial[\boldsymbol{f}_z(\boldsymbol{z})+\boldsymbol{g}_z(\boldsymbol{z})\boldsymbol{u}]}{\partial\boldsymbol{z}}|_{\boldsymbol{z}_{k-1},\boldsymbol{u}_{k-1}} \in \mathcal{R}^{n\times n}$ is the transition matrix, and $G_{k-1} = \boldsymbol{g}_z(\boldsymbol{z})|_{\boldsymbol{z}_{k-1}} \in \mathcal{R}^{n\times m}$ is the input distribution matrix

of the discretized system at time step $k-1$. The incremental form of this discrete nonlinear system can be written as

$$\Delta \boldsymbol{z}_{k+1} \approx F_{k-1}\Delta \boldsymbol{z}_k + G_{k-1}\Delta \boldsymbol{u}_k. \tag{41}$$

Hence, the dynamics of the nonlinear system can be represented by a time-varying incremental model with F_{k-1} and G_{k-1}. Since the output $\boldsymbol{y}$ is linear to $\boldsymbol{z}$, the incremental form of the output function can be written as

$$\Delta \boldsymbol{y}_{k+1} = C\Delta \boldsymbol{z}_{k+1}. \tag{42}$$

Note that when the time-scale separation principle does not hold, all the first-order terms in Eq. (41) will remain.

It has been approved that if a nonlinear system is fully observable with its output, the system output can be represented by the previous input and output measurements over a long enough time horizon $[t_{k-N}, t_k]$ [51–53]:

$$\Delta \boldsymbol{z}_{k+1} \simeq \widetilde{F}_{k-1,k-N} \cdot \Delta \boldsymbol{z}_{k-N+1} + U_N \cdot \overline{\Delta \boldsymbol{u}}_{k,k-N+1}, \tag{43}$$

$$\overline{\Delta \boldsymbol{y}}_{k,k-N+1} \simeq V_N \cdot \Delta \boldsymbol{z}_{k-N+1} + T_N \cdot \overline{\Delta \boldsymbol{u}}_{k,k-N+1}, \tag{44}$$

where symbol $\widetilde{F}_{k-a,k-b} = \prod_{i=k-a}^{k-b} F_i = F_{k-a} \cdot \ \cdots \ \cdot F_{k-b}$,

$$\overline{\Delta \boldsymbol{u}}_{k,k-N+1} = \begin{bmatrix} \Delta \boldsymbol{u}_k \\ \Delta \boldsymbol{u}_{k-1} \\ \vdots \\ \Delta \boldsymbol{u}_{k-N+1} \end{bmatrix} \in \mathcal{R}^{mN},$$

$$\overline{\Delta \boldsymbol{y}}_{k,k-N+1} = \begin{bmatrix} \Delta \boldsymbol{y}_k \\ \Delta \boldsymbol{y}_{k-1} \\ \vdots \\ \Delta \boldsymbol{y}_{k-N+1} \end{bmatrix} \in \mathcal{R}^{pN},$$

$$U_N = [G_{k-1} \ \ F_{k-1}G_{k-2} \ldots \widetilde{F}_{k-1,k-N+1} \cdot G_{k-N}] \in \mathcal{R}^{n\times mN}$$

is the controllability matrix,

$$V_N = \begin{bmatrix} C\widetilde{F}_{k-2,k-N} \\ C\widetilde{F}_{k-3,k-N} \\ \vdots \\ C \end{bmatrix} \in \mathcal{R}^{pN\times n}$$

is the observability matrix,

$$T_N = \begin{bmatrix} \mathbf{0} & CG_{k-2} & CF_{k-2}G_{k-3} & \cdots & C\widetilde{F}_{k-2,k-N+1} \cdot G_{k-N} \\ \mathbf{0} & \mathbf{0} & CG_{k-3} & \cdots & C\widetilde{F}_{k-3,k-N+1} \cdot G_{k-N} \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & CG_{k-N} \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \end{bmatrix} \in \mathcal{R}^{pN\times mN},$$

and $\mathbf{0} \in \mathcal{R}^{p\times m}$ represents a zero matrix block.

If the system is fully observable, V_N has a full column rank, and its left inverse can be obtained:

$$V_N^{left} = (V_N^T V_N)^{-1} V_N^T. \tag{45}$$

To have a full column rank V_N, N needs to satisfy $N \geq n/p$. By left-multiplying V_N^{left} to Eq. (44), and then substituting the equation of $\Delta \boldsymbol{z}_{k-N+1}$ into Eq. (43), the incremental output can be reconstructed uniquely as a function of the past input/output:

$$\begin{aligned} \Delta \boldsymbol{y}_{k+1} \simeq & C\widetilde{F}_{k-1,k-N} V_N^{left} \overline{\Delta \boldsymbol{y}}_{k,k-N+1} \\ & + (CU_N - C\widetilde{F}_{k-1,k-N} V_N^{left} T_N)\overline{\Delta \boldsymbol{u}}_{k,k-N+1}. \end{aligned} \tag{46}$$

Therefore, the output increment $\Delta \boldsymbol{y}_{k+1}$ can be reconstructed uniquely as a function of the measured input/output data of N previous steps, which can be generally rewritten as:

$$\begin{aligned} \Delta \boldsymbol{y}_{k+1} &\simeq \mathcal{F}_k \overline{\Delta \boldsymbol{y}}_{k,k-N+1} + \mathcal{G}_k \overline{\Delta \boldsymbol{u}}_{k,k-N+1} \\ &= \mathcal{F}_k \overline{\Delta \boldsymbol{y}}_{k,k-N+1} + \mathcal{G}_{k,11}\Delta \boldsymbol{u}_k + \mathcal{G}_{k,12}\overline{\Delta \boldsymbol{u}}_{k-1,k-N+1}, \end{aligned} \tag{47}$$

where $\mathcal{F}_k \in \mathcal{R}^{p\times Np}$ is the extended system transition matrix, $\mathcal{G}_k \in \mathcal{R}^{p\times Nm}$ is the extended input distribution matrix, $\mathcal{G}_{k,11} \in \mathcal{R}^{p\times m}$ and $\mathcal{G}_{k,12} \in \mathcal{R}^{p\times (N-1)m}$ are partitioned matrices from $\mathcal{G}_k$.

The direct relation between $\Delta \boldsymbol{y}_{k+1}$ and the input increment $\Delta \boldsymbol{u}_k$ can be constructed based on the online identification of $\mathcal{F}_k$ and $\mathcal{G}_k$ and the input and output measurements of N previous steps. The virtual control input $\boldsymbol{v}$ at every time step t_k can be designed to separate the first-order linear outer loop

$$\dot{\boldsymbol{y}} = \boldsymbol{v} \tag{48}$$

and the incremental form of the inner loop

$$\begin{aligned} \boldsymbol{v}\Delta t &= \Delta \boldsymbol{y}_{k+1} \\ &= \mathcal{F}_k \overline{\Delta \boldsymbol{y}}_{k,k-N+1} + \mathcal{G}_{k,11}\Delta \boldsymbol{u}_k + \mathcal{G}_{k,12}\overline{\Delta \boldsymbol{u}}_{k-1,k-N+1}, \end{aligned} \tag{49}$$

where Δt is the measurement sampling time. The virtual control input $\boldsymbol{v}$ can be designed to form a stable first-order outer linear system. And the input increment can be obtained from

$$\Delta \boldsymbol{u}_k = \hat{\mathcal{G}}_{k,11}^{-1}[\boldsymbol{v}\Delta t - \hat{\mathcal{F}}_k \overline{\Delta \boldsymbol{y}}_{k,k-N+1} - \hat{\mathcal{G}}_{k,12}\overline{\Delta \boldsymbol{u}}_{k-1,k-N+1}]. \tag{50}$$

The EINDI approach retains the advantages of INDI: they do not rely on the assumption of a perfect model. Instead, they use (extended) incremental models, which are online identified from sensor measurements, to cope with the model mismatch, system uncertainty, and other disturbances. The EINDI also possesses several advantages over INDI. EINDI does not assume that the time-scale separation principle holds, which will expand its real-world applications, such as flow divergence landings. In addition, the EINDI method allows output nonlinearities and internal dynamics ($r < n$), as long as the system is observable in Eq. (45). It does not even need to know the relative degree r, but only the number of states n for better selection of the number N. This approach does not need the measurement of full states or the calculation of the high-order differentiation of outputs, $\boldsymbol{y}^{(\boldsymbol{r})}$, but only the input and output measurements.

3.4. Extended incremental nonlinear dynamic inversion for flow divergence landings

Consider the vertical landing problem with only optical flow, and rewrite the first-order derivative of the flow divergence during vertical landings in Eq. (22) as below:

$$\dot{y} = -y^2 + \frac{1}{d}a. \tag{51}$$

The relative degree $r = 1$ and $z_1 = \xi = y$, which means there is an internal state. Complying with the condition in Eq. (13), the internal state can be chosen as $z_2 = \eta = d$. The flow divergence dynamics can be linearized around $\boldsymbol{z} = [y, d]^T$ at t_0 and be written in the incremental form:

$$\begin{aligned} \dot{y} &\approx \dot{y}_0 + \frac{\partial \dot{y}}{\partial y}|_{y_0,d_0,a_0}\Delta y + \frac{\partial \dot{y}}{\partial d}|_{y_0,d_0,a_0}\Delta d + \frac{\partial \dot{y}}{\partial a}|_{y_0,d_0,a_0}\Delta a \\ &= \dot{y}_0 - 2y_0\Delta y - a_0 d_0^{-2}\Delta d + d_0^{-1}\Delta a, \end{aligned} \tag{52}$$

where $\Delta y = y - y_0$, $\Delta d = d - d_0$, and $\Delta a = a - a_0$.

The flow divergence is the ratio of vertical velocity to height. When the height is very small, the flow divergence will change

very fast and theoretically unbounded. Therefore, the flow divergence landing process does not accept the assumption of time-scale separation, which can be proved by the time-scale analysis with linearization of the state equation [49,54] (see Appendix B). As the time-scale separation principle does not hold in flow divergence landings, the first-order terms in Eq. (52) remain.

Assuming that the above equation is a perfect model of the real system, the differentiated output can be replaced by the virtual input $\dot{y} = \nu$, and the control increment can be found by inverting Eq. (52):

$$\Delta a = d_0(\nu - \dot{y}_0 + 2y_0\Delta y + a_0 d_0^{-2}\Delta d). \tag{53}$$

Without the time-scale separation assumption, the incremental control input, obtained through the INDI routine, relies on the perfect model and the measurements of $d(t)$ and $\dot{y}(t)$. An alternative to a perfect model is the online incremental model identification, but this identification still relies on the measurement of $d(t)$, and the identified coefficients will be extremely noisy since they are partial derivatives of the noisy value $\dot{y}$.

To design the control increment with EINDI, the extended incremental model in Eq. (47) needs to be identified online with only the input and output measurements. It has been proved that the optical flow system is observable along trajectories where velocity is non-zero [55]. To have a full column rank matrix V_N, while making the number of parameters to be identified as small as possible, the smallest value which meets $N \geq n/p$ is selected: $N = 2$. The extended incremental model for flow divergence landings at any time t_k becomes

$$\begin{aligned}\Delta y_{k+1} &= \mathcal{F}_k \overline{\Delta y}_{k,k-1} + \mathcal{G}_k \overline{\Delta u}_{k,k-1} \\ &= [\overline{\Delta y}_{k,k-1}^T \ \overline{\Delta u}_{k,k-1}^T] \begin{bmatrix} \mathcal{F}_k^T \\ \mathcal{G}_k^T \end{bmatrix},\end{aligned} \tag{54}$$

where $\mathcal{F}_k \in \mathcal{R}^{1\times 2}$ is the extended system transition matrix, and $\mathcal{G}_k \in \mathcal{R}^{1\times 2}$ is the extended input distribution matrix.

The output prediction equation can be written as

$$\Delta \hat{y}_{k+1} = Z_k^T \hat{\Theta}_k, \tag{55}$$

where $Z_k^T = [\overline{\Delta y}_{k,k-1}^T \ \overline{\Delta u}_{k,k-1}^T]$ is a vector of the increments of input and output measurements at time t_k and t_{k-1}, and $\hat{\Theta}_k$ is the estimation of the parameter matrix $\Theta_k = \begin{bmatrix} \mathcal{F}_k^T \\ \mathcal{G}_k^T \end{bmatrix}$. The parameters can be online identified with the Recursive Least Square (RLS) method [52] as follows:

$$\epsilon_k = \Delta y_{k+1} - \Delta \hat{y}_{k+1}^T, \tag{56}$$

$$\hat{\Theta}_k = \hat{\Theta}_{k-1} + \frac{Cov_{k-1} Z_k}{\gamma + Z_k^T Cov_{k-1} Z_k}\epsilon_k, \tag{57}$$

$$Cov_k = \frac{1}{\gamma}\left(Cov_{k-1} - \frac{Cov_{k-1} Z_k Z_k^T Cov_{k-1}}{\gamma + Z_k^T Cov_{k-1} Z_k}\right), \tag{58}$$

where ϵ_k is the innovation term, $Cov_k \in \mathcal{R}^{(n+p)N\times(n+p)N}$ is the estimation covariance matrix, and $\gamma \in [0, 1]$ is the forgetting factor in the RLS approach for the extended incremental model identification. The RLS approach, compared to the ordinary LS method, can effectively identify the parameters of time-varying systems and also keeps these parameters relatively stable even when the excitation is not enough [52,53].

With the identified extended incremental model matrices $\hat{\mathcal{F}}_k$, $\hat{\mathcal{G}}_{k,11}$, and $\hat{\mathcal{G}}_{k,12}$ taken from $\hat{\Theta}_k$, the control input increment can be designed as

$$\Delta u_k = \hat{\mathcal{G}}_{k,11}^{-1}[\nu_k \Delta t - \hat{\mathcal{F}}_k \overline{\Delta y}_{k,k-1} - \hat{\mathcal{G}}_{k,12}\overline{\Delta u}_{k-1}], \tag{59}$$

where Δt is the measurement sampling time, and ν_k is the virtual input at time t_k. As in Eq. (23), the virtual input can be designed as a feedback linear control with a constant proportional feedback gain K_p as

$$\nu_k = -K_p(y_k - \vartheta^*). \tag{60}$$

Thus, the time response of y is an exponential decay with time constant $\tau = K_p^{-1}$.

3.5. Constant flow divergence landing simulations using EINDI

To demonstrate the feasibility of the above EINDI control law for flow divergence landing, this section presents the time response of the simulated vehicle system represented by Eqs. (20)-(21) in Fig. 2. In these numerical simulations, the forgetting factor in RLS approaches is chosen as $\gamma = 0.99$. These simulations initialize the extended transition matrices as $\mathcal{F}_0 = [1 \quad 0]$, the extended input distribution matrices $\mathcal{G}_0$ as zero matrices, and the covariance matrices as $\underline{Cov}_0 = 10\mathbf{I}$. Our preliminary simulation results show that the control performance does not sensitive to these initial values.

Fig. 2 presents three landing simulations performed with different setpoints $\vartheta^* = \{-0.1\ s^{-1}, -0.2\ s^{-1}, -0.3\ s^{-1}\}$ at the initial height of 10 m and velocity of 0 m/s. The constant proportional feedback gain for the outer linear loop in Eq. (60) is chosen to be $K_p = 1$, which affects the time constant $\tau = K_p^{-1}$ of the flow divergence response. The states of the system, such as height d and vertical velocity v, and their corresponding flow divergence outputs ϑ are shown in Fig. 2 (a) and (b), respectively.

It can be seen that the desired setpoints are tracked very quickly and consistently throughout the landings. This leads to an exponential decrease of both height and velocity to almost zero at the same time. During this maneuver, the vehicle first accelerated and then decelerated to zero velocity to touch the ground. This landing profile generated by the constant flow divergence strategy is certainly recommended, but its practical feasibility depends on the vehicle limitations, such as the maximum vertical velocity that can be achieved. These results show that EINDI can adapt quickly and achieve smooth landings with good tracking performance. However, it must be highlighted that the simulations used a perfect system with accurate flow divergence, which highly unlikely to happen in real-world conditions.

Fig. 3 presents one landing simulation performed with measurement noise (without using filters) to examine the system response in a more realistic scenario. The simulated noise is zero-mean normal distributed white noise. The standard deviation of the noise is 0.003 s^{-1}. In this figure, both height and velocity converge to zero at almost the same time. The flow divergence is tracked at the desired setpoint. However, it is expected that the tracking of flow divergence in the presence of measurement noise becomes more challenging when the vehicle is very close to the ground, e.g., $t > 12$ s in Fig. 3 (b). The main reason is that when the output is at a nearly steady state, the noise-to-signal ratio becomes more influential to the incremental model identification. This observation can also be detected during the real-world experiment, as shown in Section 4. Nevertheless, the tracking of the flow divergence is converging in the presence of measurement noise. Additionally, the landing performance is merely affected, as seen from the height and velocity.

4. Real-world flight tests

In this section, the EINDI method is implemented on an MAV and tested for constant flow divergence landings. The flight tests

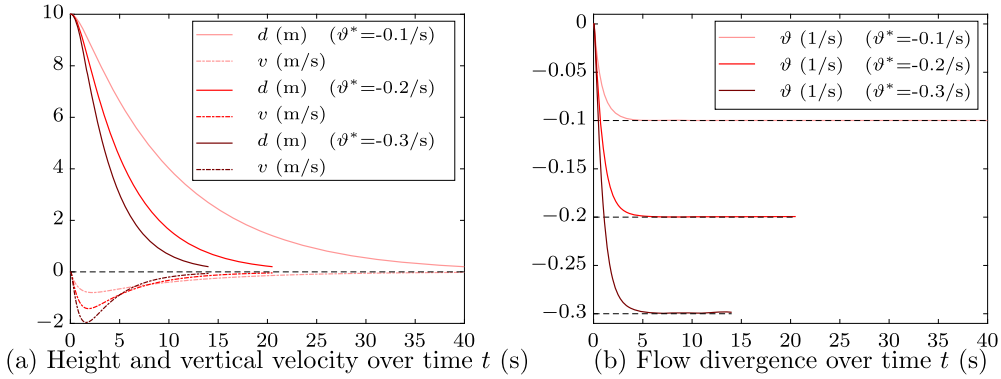

Fig. 2. Constant flow divergence landing simulations using EINDI for different desired setpoints $\vartheta^* = \{-0.1\ s^{-1}, -0.2\ s^{-1}, -0.3\ s^{-1}\}$. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

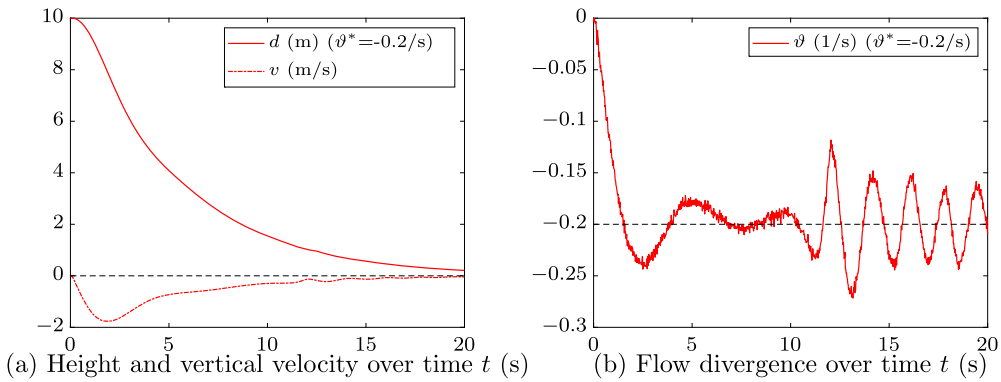

Fig. 3. Constant flow divergence landing simulations in the presence of measurement noise using EINDI for a desired setpoint $\vartheta^* = -0.2\ s^{-1}$.

are performed at different control parameters and initial conditions in a real-world outdoor environment, as shown in Fig. 4. A comparison of the flight test results using the EINDI method and the conventional feedback control is also presented and discussed in this section.

4.1. Flight test setup

The flight test setup is presented in Fig. 5. The flight tests are performed using a Parrot Bebop 2 with Paparazzi open source autopilot software[1] running on-board [56,57]. This setup replaces the original Parrot program with Paparazzi Autopilot and allows us to have direct access to the sensors and to control the MAV. The testing platform, Bebop 2, is equipped with standard sensors, such as an Inertial Measurement Unit (IMU), an altimeter, a sonar, and a Global Positioning System (GPS), and two cameras (facing forward and downward). The downward-looking camera is of particular interest to us for the landing purpose. The sonar and altimeter are activated for logging the height measurement during the flight tests to verify the landing performance.

Two software modules are created in Paparazzi Autopilot to run the respective algorithms on-board the MAV: (1) a computer vision module, and (2) a control module, as presented in Fig. 5. The computer vision module is programmed to process images captured from the camera and to execute computer vision algorithms in flight tests. For each image captured, features or corners are detected in one image using the Features from Accelerated Segment Test (FAST) algorithm [58,59] and tracked in the next image using the Lucas-Kanade tracker [60]. To estimate the flow divergence,

Fig. 4. Outdoor environment for flight tests.

Eq. (61) is used, as proved in [22]. The image distances between every two corners at one image, $\kappa_{(t-\Delta t),i}$, and at the next image, $\kappa_{t,i}$, are computed. The expansion and contraction of the flow

[1] Paparazzi Autopilot: http://wiki.paparazziuav.org.

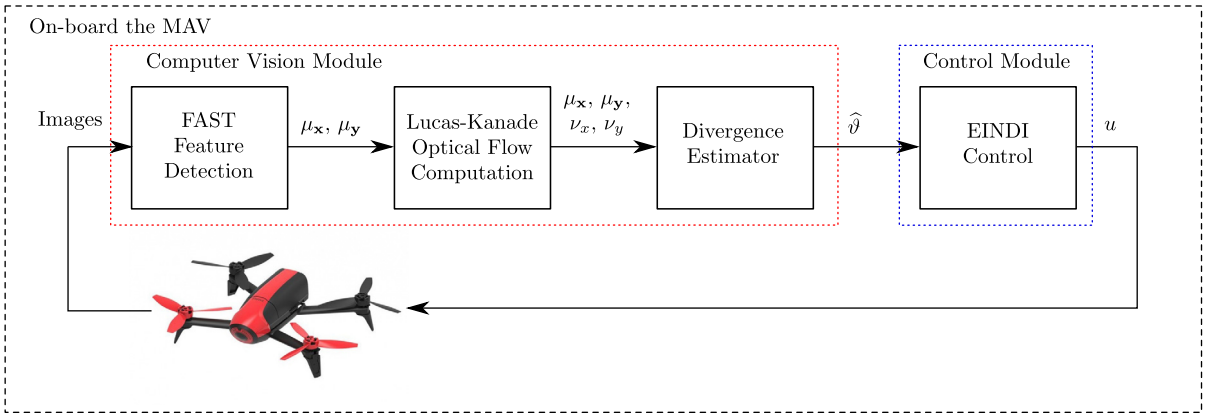

Fig. 5. Experimental setup using the Parrot Bebop 2 as the testing platform and Paparazzi open source autopilot. All the computer vision and control algorithms run on-board the MAV. In the computer vision module, FAST detects corners (μ_x, μ_y) in each image. Then, Lucas-Kanade tracks these corners in the next image and computes the optical flow (ν_x, ν_y). Lastly, the divergence estimator uses Eq. (61) to compute flow divergence ($\widehat{\vartheta}$). In the control module, the EINDI controller receives only $\widehat{\vartheta}$ as the input and commands (u) the throttle of the four motors.

can then be measured by computing the ratio of $\kappa_{(t-\Delta t),i} - \kappa_{t,i}$ to $\kappa_{(t-\Delta t),i}$. By taking the average of these ratios, and with a known time interval between these two images Δt, the flow divergence $\widehat{\vartheta}$ can be estimated using Eq. (61):

$$\widehat{\vartheta} = \frac{1}{M} \cdot \frac{1}{\Delta t} \sum_{i=1}^{M} [\frac{\kappa_{(t-\Delta t),i} - \kappa_{t,i}}{\kappa_{(t-\Delta t),i}}], \tag{61}$$

where M is the total number of tracked corners.

A low pass filter is implemented to reduce the noise in the estimation. The control module is programmed to receive the outputs from the computer vision module and command the throttle of four motors. A vertical control loop in the module executes either the EINDI or the conventional proportional feedback (for comparison purposes) controllers to perform vertical landings. The horizontal movement is stabilized using the average of translational optical flow estimates (rotational components in the optical flow are estimated and reduced using the IMU). All the flight tests begin with manually taking off and flying to a particular height and then switch to the automatic landing mode, which runs only the modules to perform landings. For the performance validation and analysis purposes, the height is obtained from the measurement of sonar if it is below or equal to 4 m and altimeter if it is higher than 4 m due to the limitation and accuracy of the sensors.

4.2. Constant flow divergence landings with the EINDI and conventional feedback control

In this subsection, the landing performance is compared between the conventional feedback control and the newly designed EINDI. To examine the stability, tracking performance, and adaptability of the controllers, the automatic landing is started at a height near to the ground, e.g., 3 m. Landings at this height impose a challenge on the controller keeping a constant flow divergence within a short period of time.

Most conventional approaches use a straightforward proportional controller to regulate flow divergence to a desired constant for the landings [14–17]. It is commonly known that this controller has a fast response time in most ideal cases as the command is directly influenced by the system response, i.e., $u_p = -K_p(\vartheta - \vartheta^*)$. To examine the stability and performance of using such linear controllers on a nonlinear system, several MAV landing tests are conducted by tracking a constant flow divergence of $\vartheta^* = -0.1\ s^{-1}$. Fig. 6 shows the results of three constant flow divergence landings using the conventional feedback control at different control gains K_p. Each row in Fig. 6 presents the MAV height, vertical velocity, and flow divergence logged during the flight tests for a different $K_p = \{0.03,\ 0.3,\ 0.7\}$ (from the top to the bottom). It is clearly seen from the flow divergence shown in Fig. 6 (a) that for a small control gain ($K_p = 0.03$) the tracking of flow divergence is unsuccessful as the flow divergence deviates away from the desired tracking setpoint. For a fine-tuned control gain ($K_p = 0.3$), the tracking performance is good, except when the vehicle is close to the ground, as presented in Fig. 6 (b). For a larger control gain ($K_p = 0.7$), oscillations occur right from the beginning, which is reflected on the scale of the flow divergence, as shown in Fig. 6 (c). The oscillations become larger when it is close to the ground, which can be observed from their height measurements. These flight test results show that the conventional feedback control requires a proper gain selection for achieving stable and good landing profiles. Additionally, these control gains also depend on the height and velocity and need to be estimated and scheduled accordingly [21,22].

In contrast, the proposed EINDI method does not require a complicated gain selection or scheduling. Here, we would like to examine its stability and performance during landings (or how fast the control method can adapt to achieve stable landings). Fig. 7 shows the MAV height, vertical velocity, and flow divergence during the flight test using EINDI. It can be clearly seen that the vertical control using the EINDI method achieves a high-performance and smooth landing profile as it tracks the setpoint immediately and steadily during the flight. Additionally, if the setpoint ϑ^* is compared with the estimated divergence ϑ, the Root Mean Square Error (RMSE) of the tracking is smaller, i.e., 0.10 s^{-1}, compared to the conventional feedback control (0.58 s^{-1}, 0.18 s^{-1}, 0.18 s^{-1} for $K_p = \{0.03,\ 0.3,\ 0.7\}$, respectively). Despite showing a fast and better tracking performance, no oscillation occurs during the landing using EINDI, as this can be observed from the height measurement.

4.3. Fast landing with larger desired divergence setpoints

To achieve fast constant flow landing maneuvers, it requires not only accurate estimates of the flow divergence, but also the quick adaptation of the controller. In addition, the practical feasibility of these maneuvers depends on the vehicle limitations, such as the maximum achievable vertical velocity. For verifying fast landing maneuvers, larger divergence setpoints are assigned and tracked using the EINDI method for the vertical control. Fig. 8 presents the constant flow divergence landings using EINDI with different divergence setpoints $\vartheta^* = \{-0.1\ s^{-1},\ -0.2\ s^{-1},\ -0.3\ s^{-1}\}$.

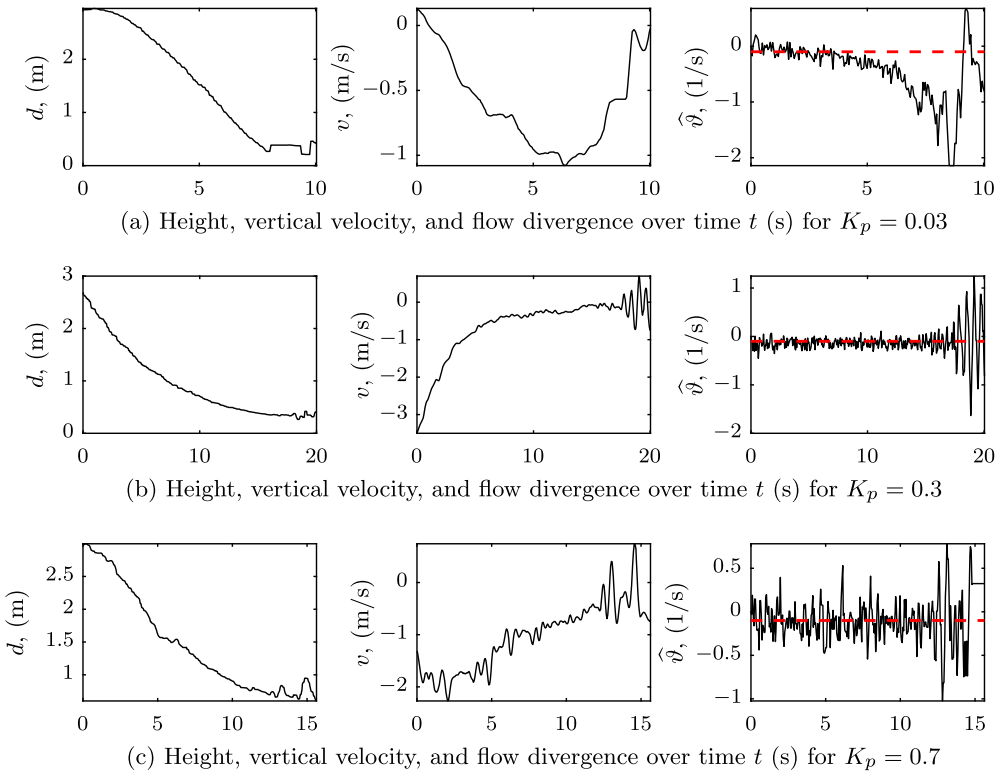

Fig. 6. Constant flow divergence landings ($\vartheta^* = -0.1\ s^{-1}$ indicated by the dashed line) using the conventional feedback control with different control gains $K_p = \{0.03,\ 0.3,\ 0.7\}$. (a) Small control gains: Unsuccessful tracking of the flow divergence, (b) Fine-tuned control gains: Reasonable performance with small oscillations when it is close to the ground, and (c) Large control gains: Oscillations occur right from the beginning and become larger when it is near to the ground.

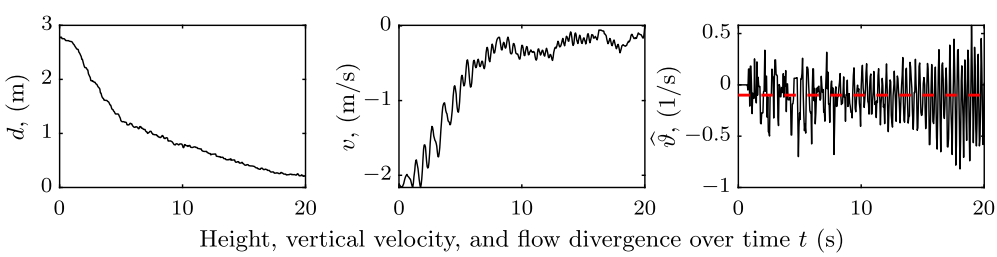

Fig. 7. Constant flow divergence landings ($\vartheta^* = -0.1\ s^{-1}$ indicated by the dashed line) using EINDI.

Interestingly, the proposed EINDI method can track these large divergence setpoints very well. All the landings are successful with the exponential decreases in both their velocity and height to almost zero at the same time.

4.4. Flexible landing with different setpoints and initial conditions

Besides fast landings, it is also crucial to ensure that the landings can be performed at different initial conditions, such as different starting heights and velocities. For this reason, the experiments begin with a different initial height while slowly climbing (positive velocities since the previous results in Figs. 7 and 8 have presented approximately zero or negative initial velocities). Additionally, different setpoints are tracked for the landings. Fig. 9 presents the constant flow divergence landings at a height of $d \approx 12\ m$ and different initial velocities for tracking different divergence setpoints $\vartheta^* = \{-0.1\ s^{-1},\ -0.2\ s^{-1},\ -0.3\ s^{-1}\}$. As mentioned above, for validation purposes, the height measurement is obtained automatically during the flight tests from both the altimeter and sonar. Although the switching between these two sensor readings can be clearly seen at approximately 4 m, the landing trajectories are actually smooth and stable during the flight tests. Additionally, despite at heights, the EINDI controller tracks the different desired flow divergence setpoints well, and no oscillation occurs.

Based on all these flight test results, we have validated that the proposed EINDI outperforms the conventional feedback control, as it does not require an extensive gain selection and scheduling based on height estimates, yet it can guarantee a stable and desired landing profile. Additionally, it has been shown that the proposed EINDI can track large flow divergence setpoints very quickly and accurately for achieving fast and safe landings, as these setpoints were well kept constant leading to both their velocity and height decrease exponentially to almost zero at the same time. Besides, the robustness of the proposed EINDI has been tested with different initial conditions and setpoints. The results have shown that the landing trajectories are smooth and stable with good tracking performance during the flight tests. It is also important to highlight that all these flight tests presented in this paper are conducted in outdoor environments, where slight perturbation

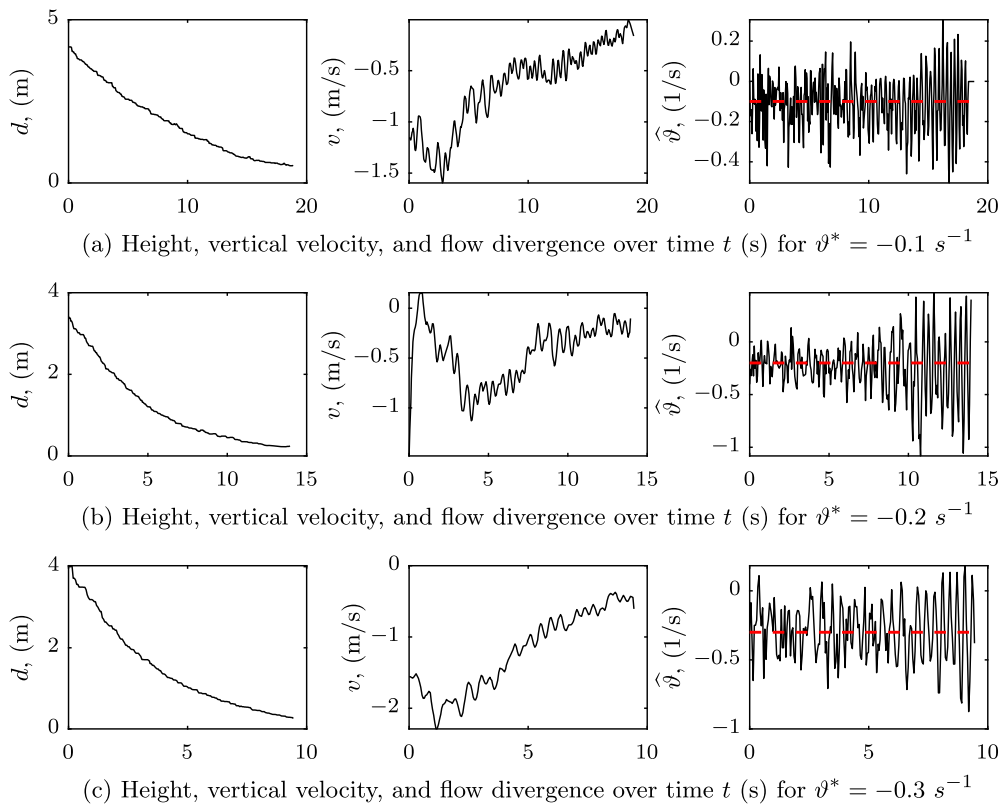

Fig. 8. Constant flow divergence landings using the EINDI method at a height of $d \approx 4\ m$ in an outdoor environment. (a) $\vartheta^* = -0.1\ s^{-1}$. (b) $\vartheta^* = -0.2\ s^{-1}$. (c) $\vartheta^* = -0.3\ s^{-1}$.

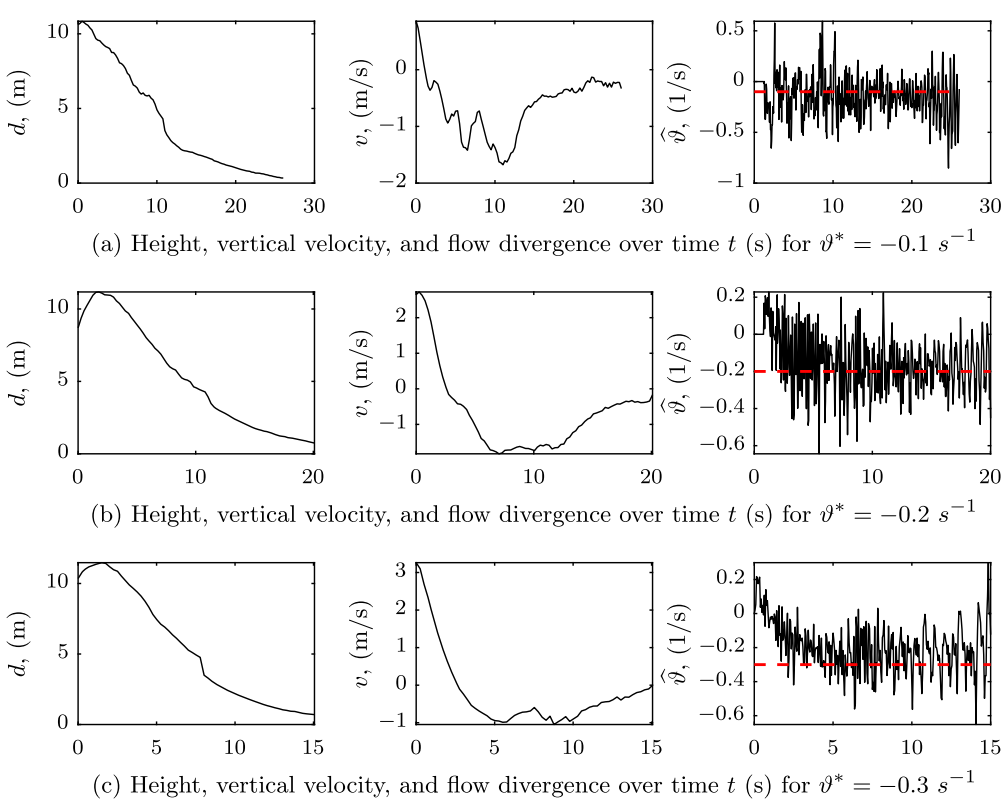

Fig. 9. Constant flow divergence landings using the EINDI method at a height of $d \approx 12\ m$ in an outdoor environment. (a) $\vartheta^* = -0.1\ s^{-1}$. (b) $\vartheta^* = -0.2\ s^{-1}$. (c) $\vartheta^* = -0.3\ s^{-1}$.

exists due to wind. However, the controller is sufficiently robust to deal with real-world conditions.

5. Conclusion

This paper formulated a novel Extended Incremental Nonlinear Dynamic Inversion (EINDI) method, based on the concept of NDI and INDI, to solve nonlinear control problems without knowing their models or full state measurements. The EINDI method retains the advantages of INDI as they do not rely on the system model. Instead, they utilize an online identified (extended) incremental model to cope with system uncertainties, model mismatch, or other disturbances. This paper showed that EINDI possesses several advantages over INDI: 1) EINDI does not rely on the time-scale separation assumption; 2) it allows output nonlinearities and internal dynamics to exist as long as the system is observable; 3) it does not even need to know the relative degree, but only the number of states for a better selection of the extended incremental model structure; and 4) it does not need the full state measurements or even the calculation of the high-order differentiation of outputs, but only the input and output measurements. These traits will expand its real-world applications with output nonlinearities, such as optical flow control problems.

The EINDI method allows MAVs to perform constant divergence landings, without estimating or measuring distances with additional sensors. This is the first time that the INDI control law is generalized to be applied to optical flow-based control problems, where the time-scale separation principle does not hold and the full state measurements are unavailable. The proposed method has been applied to the constant flow divergence landing and verified in real-world flight tests. The results show that 1) the tracking performance of EINDI is better than the conventional feedback control, 2) the landing trajectories are smooth without any oscillation, and 3) the adaptation of EINDI is fast even for landings at different heights and desired setpoints. Further investigation and experimentation are recommended into multiple-input and multiple-output nonlinear control problems. Additionally, the EINDI-based constant flow divergence strategy could also be applied to achieve autonomous navigation tasks, such as obstacle avoidance, with a forward-looking camera. It is expected that this elegant control strategy will allow a stable forward flight with a high initial velocity which will exponentially decrease with the distance to an object and will eventually become zero when approaching the object. Therefore, MAVs could navigate safely without adding more payloads or computations, which could unlock the potential of MAVs in real-world applications.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Time scale analysis for conventional vertical landing model

The equations of motion of the vertical landing model in Eq. (20) can be rewritten in the linear state space form as below:

$$\dot{\boldsymbol{x}} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \boldsymbol{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} a, \tag{A.1}$$

where $\boldsymbol{x} = [d, v]^T$, and a is the vertical acceleration. Taking the actual vertical thrust T into this model, we will have

$$a = \frac{T}{m_a} - g, \tag{A.2}$$

where m_a is the mass, g is gravity, and the drag force or the wind disturbances are not counted in this model. If the actuators can be approximated with a first-order dynamical equation, we will have

$$\frac{T(s)}{T_{in}(s)} = \frac{K\eta}{s+\eta}, \tag{A.3}$$

where T_{in} is the input of the actuator, which is an electrical signal, K is the gain of the actuator, which is often Newton/volt, and η represents the fastness of the actuator dynamics. Therefore, the dynamics of a becomes

$$\dot{a} = \frac{\dot{T}}{m_a} = -\eta a - \eta g + \frac{\eta K}{m_a} T_{in}. \tag{A.4}$$

This can be transformed to the singularly perturbed form by picking $\epsilon_1 = \eta^{-1}$:

$$\begin{aligned} \dot{\boldsymbol{x}} &= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \boldsymbol{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} a \\ \epsilon_1 \dot{a} &= -a - g + \frac{K}{m_a} T_{in}, \end{aligned} \tag{A.5}$$

where the actuator dynamics is sufficiently fast and $\epsilon_1 \ll 1$. Therefore, $\mathbf{x}$ is the slow state and a is the fast state, and the time-scale separation principle holds for the incremental model simplification as in Eq. (28).

Appendix B. Time scale analysis for flow divergence landing model

The first-order derivative of the flow divergence in Eq. (22) is rewritten as below:

$$\dot{y} = -y^2 + \frac{1}{d} a, \tag{B.1}$$

where $y = v \cdot d^{-1}$ is the observed flow divergence, and a is the vertical acceleration. The vertical velocity v is bounded due to the vehicle limitations, but the height d can be near zero, which makes the flow divergence theoretically unbounded.

To investigate the details, another approach will be used to better analyze the time scale of the new nonlinear system dynamics in z coordinate. This approach will linearize the nonlinear dynamical equation, Eq. (B.1), around an operating point, and then examine the eigenvalues of the linearized dynamics. The eigenvalues will represent the fastness of the corresponding system modes [54,49].

Complying with the condition in Eq. (13), the new $\boldsymbol{z}$ state can be chosen as $\boldsymbol{z} = [y\ d]^T$ as discussed in subsection 3.4. The height d is a slow state compared to a when the actuator dynamics is fast. The rest content will analyze the dynamics of y and a by assuming that the height d can be seen as a constant d_0 around the operating point at t_0. Therefore, the flow divergence dynamics in Eq. (B.1) can be linearized by using the first-order Taylor series expansion around $[y_0\ a_0]^T$:

$$\begin{aligned}\dot{y} \approx & \dot{y}_0 + \frac{\partial(-y^2 + d_0^{-1}a)}{\partial y}|_{y_0,a_0}(y - y_0) \\ & + \frac{\partial(d_0^{-1}a)}{\partial a}|_{y_0,a_0}(a - a_0) \\ = & \dot{y}_0 - 2y_0(y - y_0) + d_0^{-1}(a - a_0) \\ = & -2y_0 y + d_0^{-1}a + (\dot{y}_0 + 2y_0^2 - d_0^{-1}a_0).\end{aligned} \tag{B.2}$$

Since the dynamics of a in Eq. (A.4) is linear, the linearized equation of motion of $[y\ a]^T$ around $[y_0\ a_0]^T$ can be written in the linear state space form as follows:

$$\begin{bmatrix} \dot{y} \\ \dot{a} \end{bmatrix} = \begin{bmatrix} -2y_0 & d_c^{-1} \\ 0 & -\eta \end{bmatrix} \begin{bmatrix} y \\ a \end{bmatrix} + \begin{bmatrix} \dot{y}_0 + 2y_0^2 - d_c^{-1}a_0 \\ -\eta g + \eta K m_a^{-1} T_{in} \end{bmatrix}. \tag{B.3}$$

The eigenvalues of the system matrix are

$$\begin{aligned}\lambda_1 &= -2y_0, \\ \lambda_2 &= -\eta,\end{aligned} \tag{B.4}$$

where λ_2 is a constant value determined by the actuator performance, and $\lambda_1 = -2y_0 = -2v(t_0)d(t_0)^{-1}$ theoretically can be any value in $(-\infty, +\infty)$. Therefore, the time-scale separation assumption does not hold for $[y\ a]^T$ during the flow divergence landing process.

Note that this linearization assumes only y and a are variables because we know that d is slower than a and only tries to compare the fastness of y and a. Even if d is considered as a variable in the analysis, we will have the same conclusion that the time-scale separation assumption does not hold because the eigenvalues are not all bounded.

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